

ODISHA SPATIAL SCHOOL EDUCATION MASTERPLAN (2026–2031)

Audited Operations Research & PWD Hill Cost Calibrated Strategy Across 314 CD Blocks

59.1% → 91.6% Secondary Coverage	₹6,008.4 Cr Hill-Adjusted Outlay	314 Blocks 30 Districts Analyzed	1.8x GSDP ROI ₹10,796 Cr Value	9,144 Posts Teachers (25% Hardship)
-------------------------------------	-------------------------------------	-------------------------------------	-----------------------------------	--

1. Strategic Context & Operations Research Methodology

- **Grade 8 → 9 Spatial Dropout Cliff:** While primary access achieves >74%, secondary coverage drops to 59.1% due to distance barriers (mean distance >6.8 km in Eastern Ghats), driving >42% tribal dropout with girls dropping out at 2.3x higher rates without safe transit.
- **PuLP MILP & PWD Hill Cost Multipliers:** Interventions are optimized using PuLP Mixed-Integer Linear Programming (MCLP) under capital constraints, incorporating **Tobler's Hiking Slope Resistance (1.8x–2.4x)** and **PWD Hill Cost Multipliers (+18% to +30%)** to model realistic ghat logistics.

2. Statewide Multi-Tier Investment & Physical Targets (Hill Cost Calibrated)

Education Tier	Norm	Existing	Baseline	Upgrades	New Campuses	Transit Hubs	Target Access	Total Est. Outlay
Primary	1.0 km	37,639	74.9%	1,333	730	719	99.2%	₹1,030.0 Cr
Upper Primary	3.0 km	13,923	67.9%	1,726	934	899	99.2%	₹2,399.5 Cr
Secondary	5.0 km	6,539	59.1%	2,182	1,195	1,172	91.6%	₹6,008.4 Cr
Higher Secondary	7.0 km	1,943	48.7%	2,819	1,495	1,407	81.2%	₹13,662.1 Cr

3. 314 CD Blocks Regional Stratification, Transit Fleet & Teacher Cadre

- **High Vulnerability (72 Blocks in 9 Tribal Dists):** Coverage <45%. Priority for dedicated girls' hostels, feeder transit, and teacher hardship pay.
- **Central & Agrarian (142 Blocks):** Coverage 55–70%. Focus on expanding existing Upper Primary schools into High Schools.
- **Coastal & Deltaic (100 Blocks):** Coverage >75%. Focus on STEM infrastructure and 396 cyclone-resilient structural retrofits.

588 Girls' Hostels Guarantees female retention	1,027 Mini-Buses (24-Seater) @ ₹4.80L/yr Opex	603 Feeder Vans (12-Seater) @ ₹3.00L/yr Opex	₹67.6 Cr Annual Opex Fuel, repairs & chaperones	9,144 Subject Teachers 25% Hardship Allowance
---	--	---	--	--

4. 5-Year Capital Rollout, Calibrated Economic ROI & Cabinet Directives

Phase	Target Corridors	Outlay	Upgrades	New Campuses	Transit Hubs	Access Gain
Phase 1 (Y1–Y2)	High Vulnerability Tribal Corridors (9 Dists)	₹2,704 Cr (45%)	981	537	644	+16.5%
Phase 2 (Y3–Y4)	Mineral Belts & Western Plateaus (11 Dists)	₹2,103 Cr (35%)	763	418	351	+11.2%
Phase 3 (Y5)	Coastal Deltas & Cyclone Retrofits (10 Dists)	₹1,202 Cr (20%)	436	239	175	+4.8%

Audited Economic ROI: Retaining 184,000 secondary students generates **₹10,796.3 Cr lifetime GSDP value** (0.75x informal labor discount), delivering a **1.8x Benefit-Cost Ratio** on **₹6,008.4 Cr** capital outlay.

Immediate Directives: **1. Tribal Teacher Cadre:** Sanction 9,144 posts with 25% Remote Area Allowance & 3-yr bond. **2. PESA Land Fast-Track:** Transfer revenue wasteland via Gram Sabha consent (PESA Sec 4i). **3. SHG Transit Operations:** Partner with Mission Shakti SHGs for bus/feeder chaperones. **4. GIS-Linked Disbursals:** Require GPS validation in solver prior to fund release.